GaussNiche 5-D downstream HSM — full report (RND / buffer-out / uniform+)

Median per metric x sampler — pc5 predictors

metric	RND	buffer-out	uniform+
COR (truth)	0.759	0.725	0.618
RMSE (truth)	0.222	0.275	0.321
AUC	0.898	0.970	0.970
CBI	0.842	0.822	0.796
TSS	0.722	0.867	0.875
Sensitivity	0.933	0.956	0.967
Specificity	0.811	0.929	0.922
Kappa	0.722	0.867	0.875
COR	0.718	0.864	0.870
R2	0.526	0.750	0.764
RMSE	0.354	0.271	0.273
Deviance	135.359	82.380	79.866

Median per metric x sampler — raw12 predictors

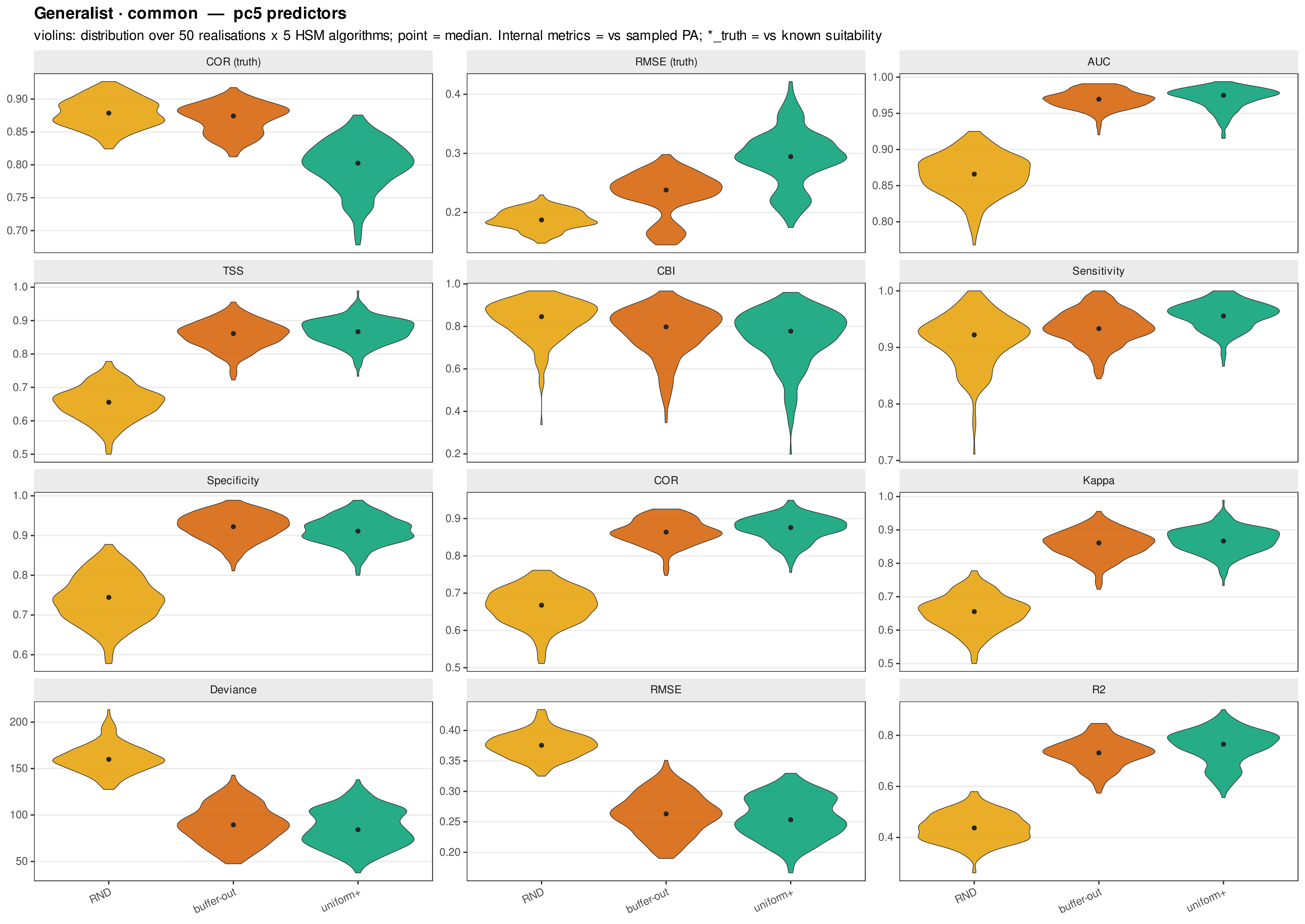
metric	RND	buffer-out	uniform+
COR (truth)	0.730	0.687	0.557
RMSE (truth)	0.237	0.297	0.374
AUC	0.887	0.964	0.964
CBI	0.849	0.832	0.786
TSS	0.700	0.856	0.857
Sensitivity	0.931	0.944	0.956
Specificity	0.800	0.922	0.911
Kappa	0.700	0.856	0.857
COR	0.700	0.851	0.854
R2	0.522	0.746	0.765
RMSE	0.366	0.284	0.289
Deviance	145.421	91.746	88.804

Internal metrics = held-out test vs sampled PA; *_truth = predicted map vs known suitability.

Where uniform+ beats the naive samplers — pc5 predictors

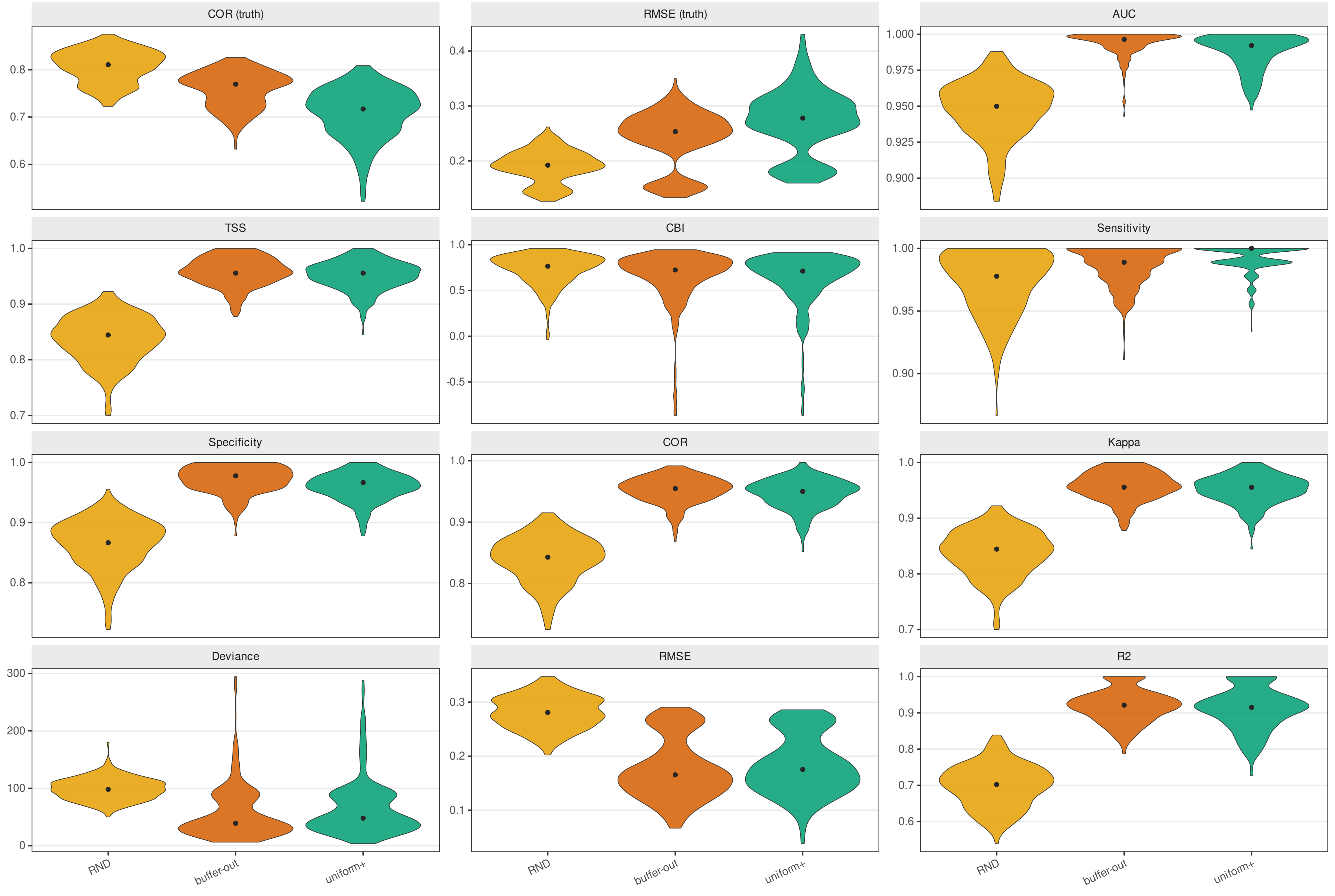
cell = margin of uniform+ vs the BETTER naive sampler (>0 better); median over algos x realisations


Where uniform+ beats the naive samplers — raw12 predictors
cell = margin of uniform+ vs the BETTER naive sampler (>0 better); median over algos x realisations
Generalist · common
Specialist · common
Generalist · rare
Specialist · rare
COR (truth)
-0.118
-0.136
-0.160
-0.071
RMSE (truth)
-0.140
-0.112
-0.132
-0.063
AUC
+0.006
-0.005
+0.009
-0.021
CBI
-0.114
-0.058
-0.007
-0.064
TSS
+0.011
-0.011
+0.011
-0.051
Sensitivity
+0.022
+0.000
+0.011
+0.000
Specificity
+0.000
-0.006
+0.000
-0.056
Kappa
+0.011
-0.011
+0.011
-0.051
COR
+0.018
-0.004
+0.014
-0.034
R2
+0.040
-0.005
+0.023
-0.046
RMSE
+0.014
-0.005
+0.008
-0.011
Deviance
+6.117
-11.611
+5.128
-1.204
uniform+ vs naive
(RND & buffer-out)
beats both
beats one
beats none


Generalist · common — pc5 predictors
violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability
COR (truth)
RMSE (truth)
AUC
TSS
CBI
Sensitivity
Specificity
COR
Kappa
Deviance
RMSE
R2
RND
buffer-out
uniform+

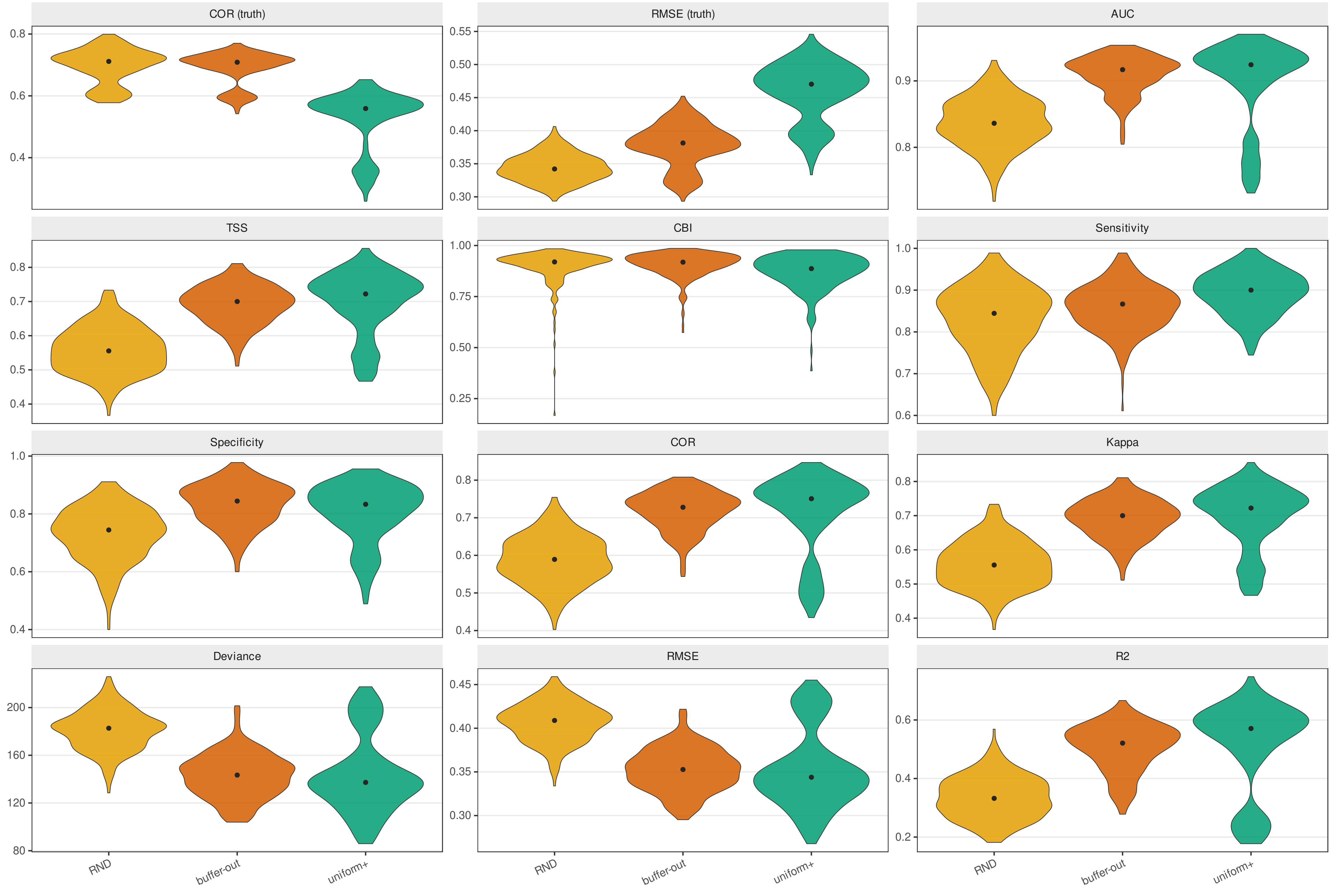
Specialist · common — pc5 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



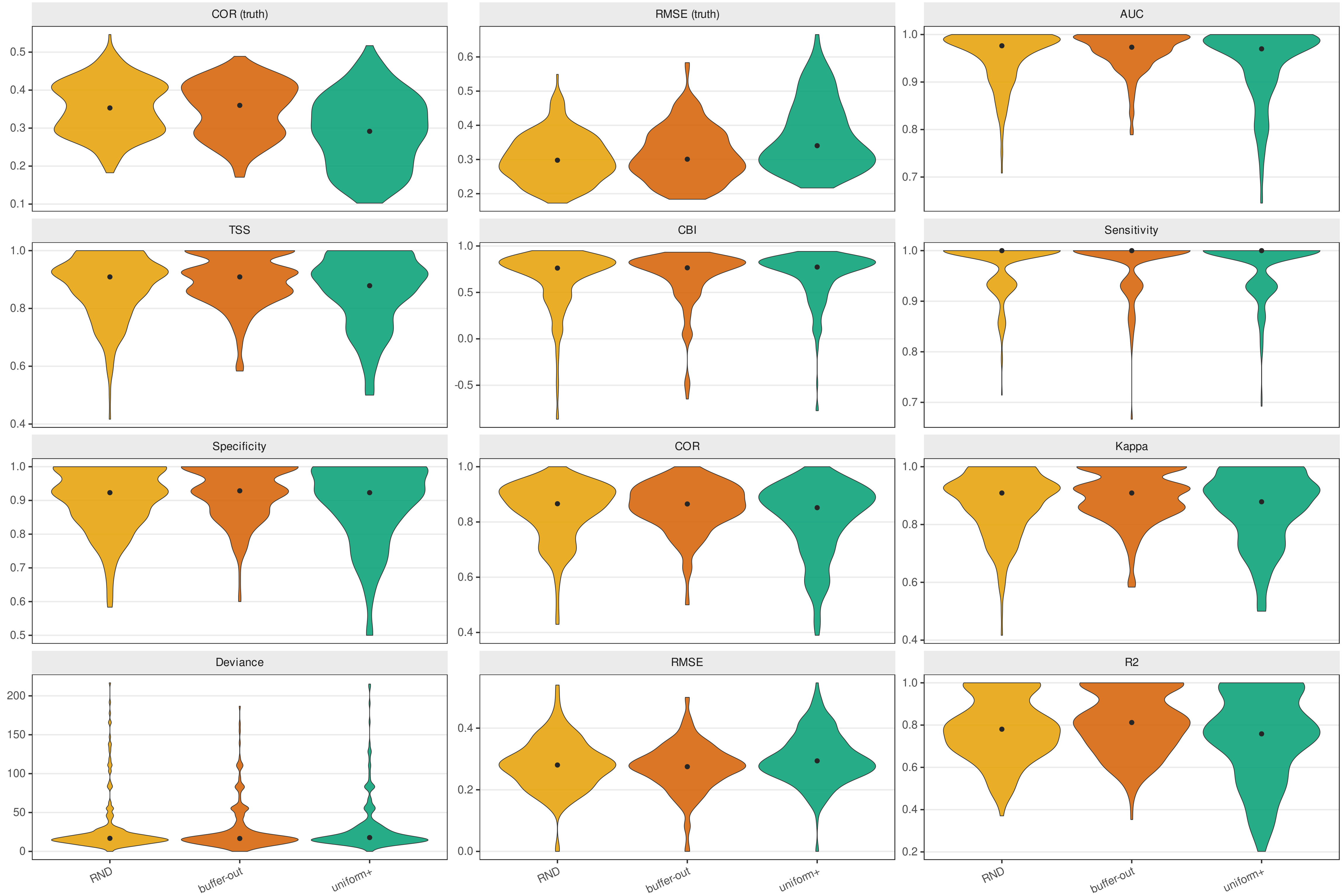
Generalist · rare — pc5 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



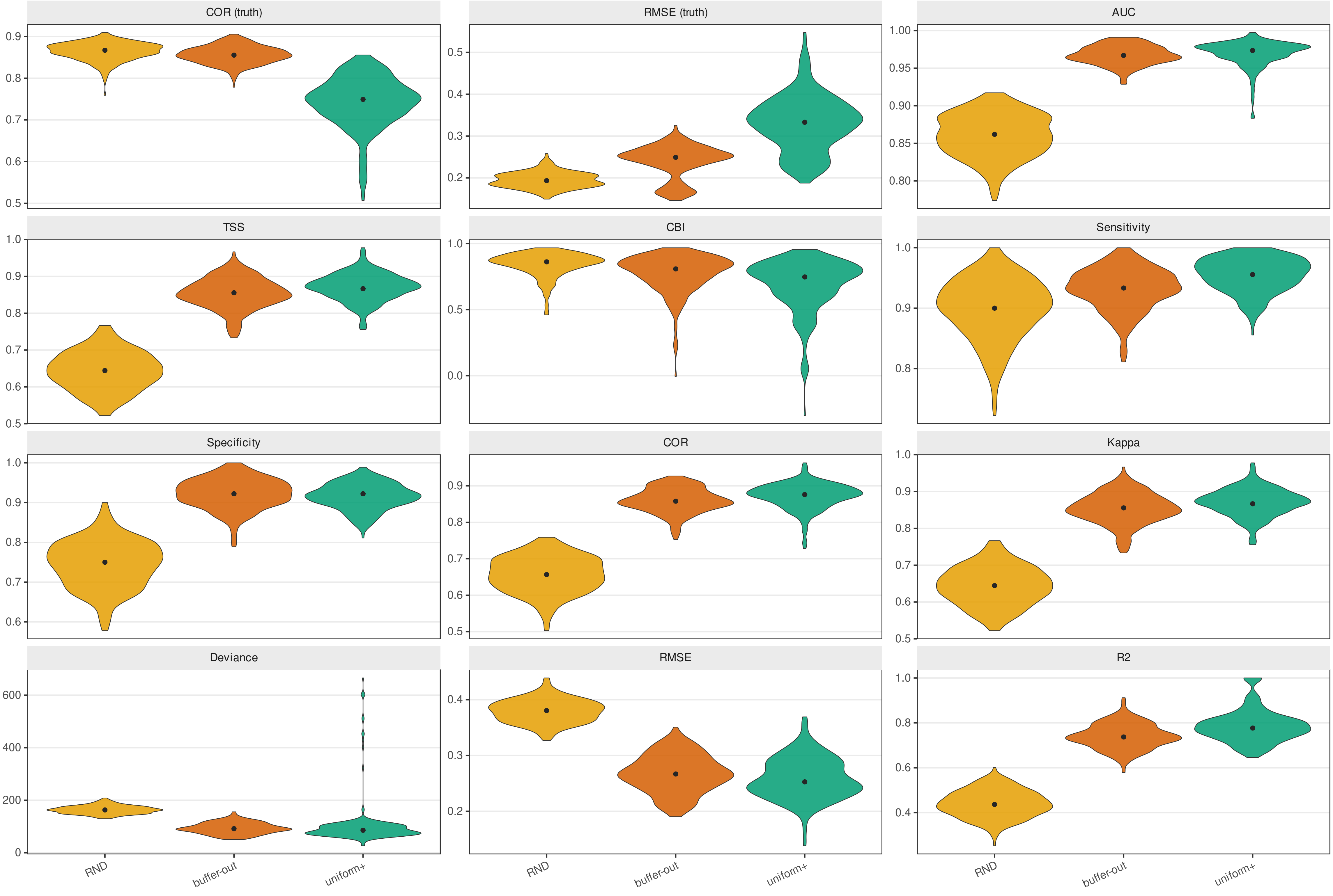
Specialist · rare — pc5 predictors

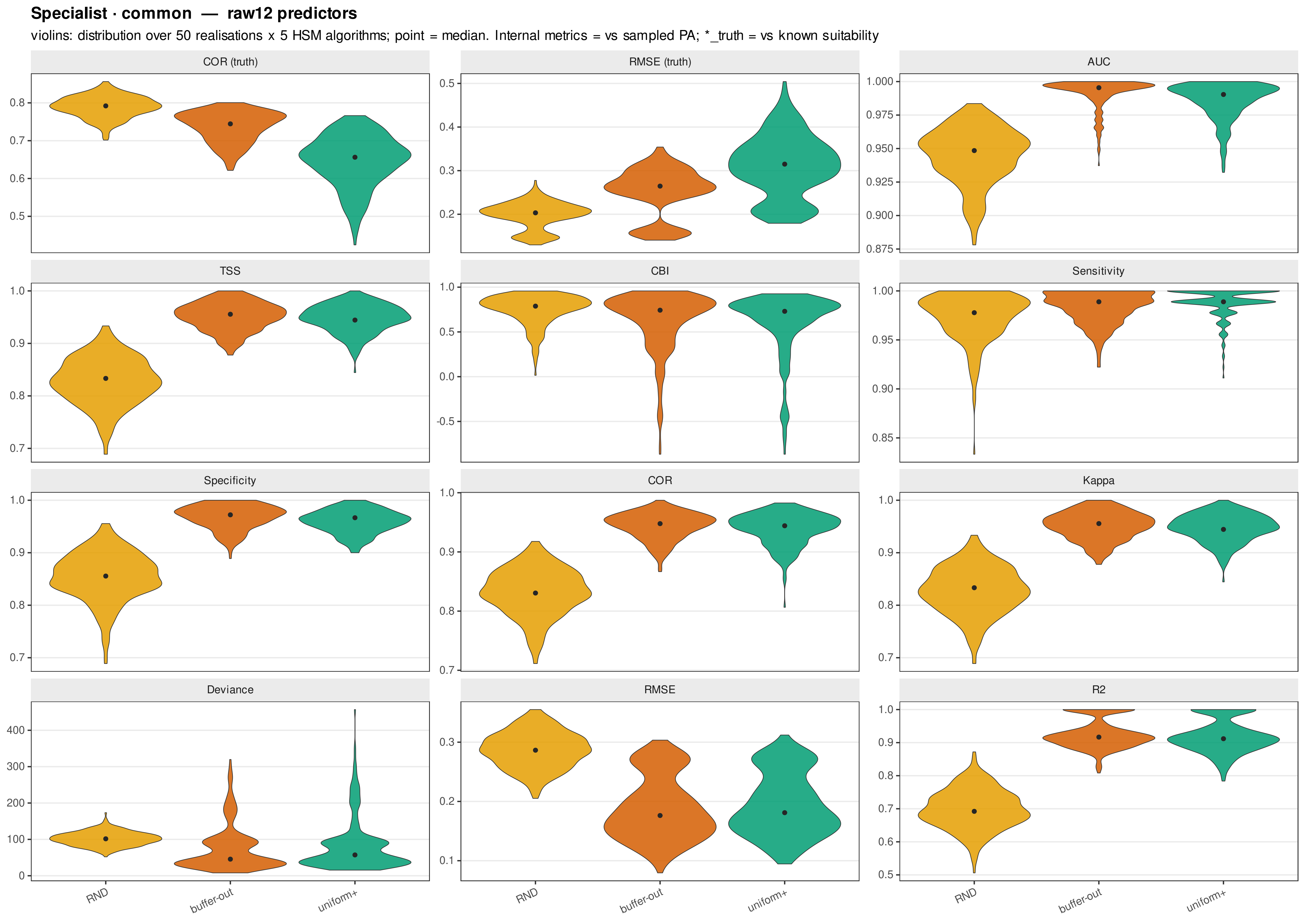
violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



Generalist · common — raw12 predictors

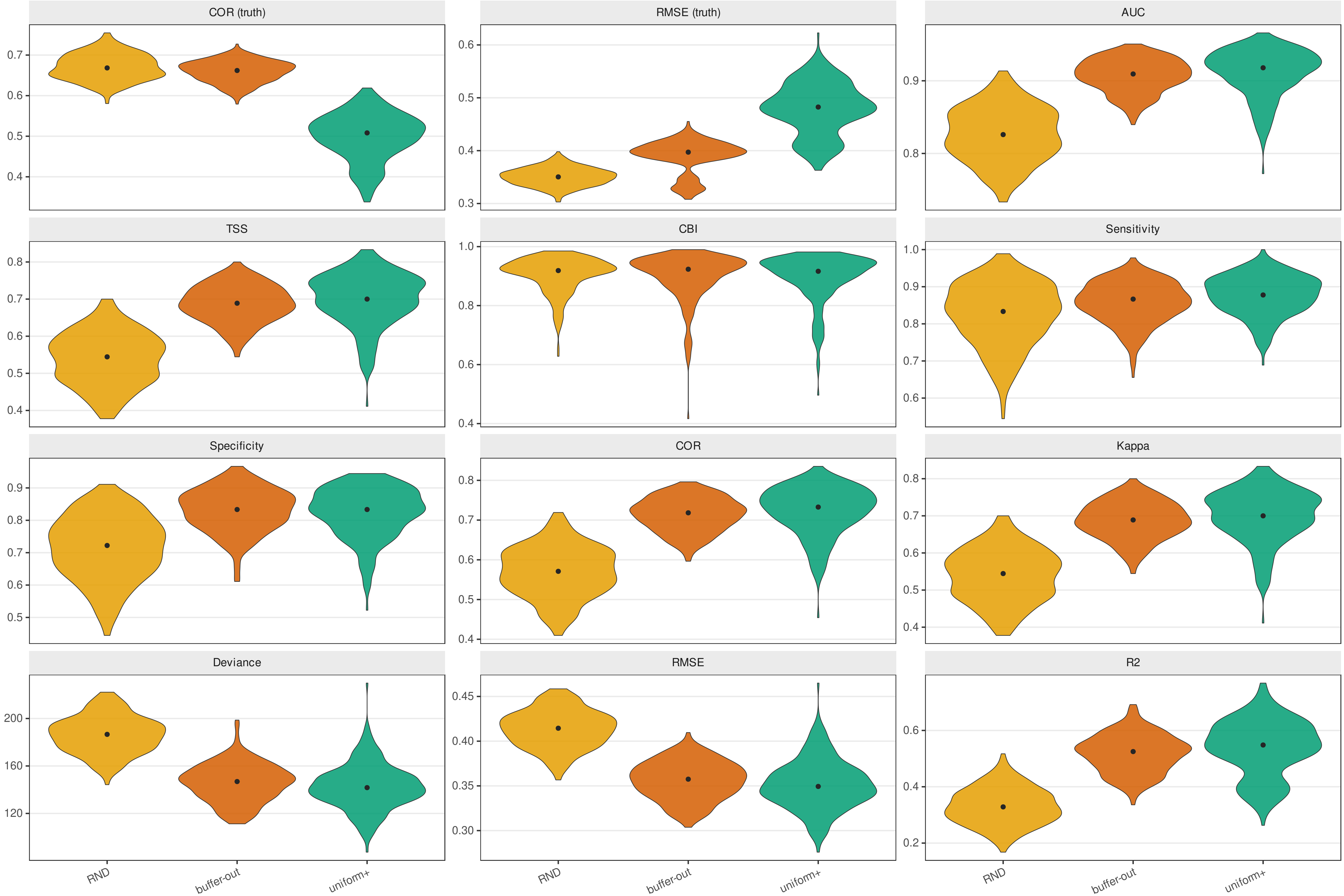
violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability




Specialist · common — raw12 predictors
violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability
COR (truth)
RMSE (truth)
AUC
TSS
CBI
Sensitivity
Specificity
COR
Kappa
Deviance
RMSE
R2
RND
buffer-out
uniform+

Generalist · rare — raw12 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability



Specialist · rare — raw12 predictors

violins: distribution over 50 realisations x 5 HSM algorithms; point = median. Internal metrics = vs sampled PA; *_truth = vs known suitability